Unit: 6 Unit 8: Quadratics
Topic: Vertex Form and Intercept Form

Name: ______________
Date: __________

Advanced (Habanero)

Q1. What is the vertex form of a quadratic function?
(A) $y = a(x-h)^2 + k$
(B) $y = ax^2 + bx + c$
(C) $y = a(x+h)^2 - k$
(D) $y = x^2 + bx + c$
(E) $y = (x-h)^2 + k$

Q2. Which of the following is the intercept form of $y = x^2 - 4x - 5$?
(A) $y = (x-2)(x+1)$
(B) $y = (x+2)(x-1)$
(C) $y = x(x-5)$
(D) $y = (x+5)(x-1)$
(E) $y = x(x+5)$

Q3. What is the value of a in the vertex form $y = a(x-2)^2 + 1$ if the graph passes through $(0, 5)$?
(A) 2
(B) 1
(C) -2
(D) -1
(E) 4

Q4. What is the y-intercept of the graph of $y = (x+1)(x-3)$?
(A) -3
(B) 3
(C) -2
(D) 2
(E) 0

Q5. Which of the following quadratic functions has its vertex at $(2, 3)$?
(A) $y = x^2 - 4x + 5$
(B) $y = (x-2)^2 + 3$
(C) $y = x^2 + 4x + 5$
(D) $y = (x+2)^2 - 3$
(E) $y = x^2 - 2x + 1$

Q6. What are the x-intercepts of $y = x^2 + 2x - 3$?
(A) $x = -3, 1$
(B) $x = -1, 3$
(C) $x = -2, 2$
(D) $x = 1, -1$
(E) $x = -1, -3$

Q7. What is the axis of symmetry of the graph of $y = x^2 - 4x + 3$?
(A) $x = 2$
(B) $x = -2$
(C) $x = 1$
(D) $x = -1$
(E) $x = 4$

Q8. Which quadratic function has its vertex at $(-2, 1)$ and y-intercept at $(0, 3)$?
(A) $y = (x+2)^2 - 2$
(B) $y = (x-2)^2 + 1$
(C) $y = x^2 + 4x + 3$
(D) $y = x^2 - 4x + 3$
(E) $y = (x+2)^2 + 2$

Open Ended Questions (FRQ)

Q9. Write the quadratic function $y = x^2 - 6x + 5$ in vertex form. Show your work and explain the steps.

Q10. Graph the quadratic function $y = (x-1)(x+3)$ and identify its x-intercepts, y-intercept, and axis of symmetry.

Chef's Tips

- Tip 1: Encourage students to check their work by plugging in values.
- Tip 2: Remind students that the vertex form of a quadratic function can be written as $y = a(x-h)^2 + k$.
- Tip 3: Suggest using graphing technology to visualize the graphs of the quadratic functions.